

Task 6

Constructing and Evaluating Synthetic Audio

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Synthetic media disclosure: The submitted recordings contain AI-generated speech. This report documents the supplied artifacts and distinguishes observed findings from unrecorded production details.

Purpose and principal finding

I generated two audio files using ElevenLabs and supplied an AI-detection screenshot for this assessment. These recordings are the primary artifacts. Their measured lengths are 55.88 and 58.12 seconds. The supplied detector screenshot labels the analyzed recording AI-generated, with a displayed AI score of 95 percent. This agrees with my account that the audio was generated, but it does not establish detector accuracy across other recordings.

The project also retains an earlier AI-assisted Flite experiment as a documented baseline. Its settings, listening feedback, and metadata-conversion results are historical evidence and must not be attributed to my ElevenLabs exports. The written analysis and packaging were revised with AI assistance.

Artifact inventory and technical checks

Recording 01 is `AI_generated_ElevenLabs_Roger_01.mp3`. Its original export name contains `2026-10-02T00_27_09` and `Roger - Laid-Back, Casual, Resonant`. The file is 910662 bytes and 55.875875 seconds long.

Recording 02 is `AI_generated_ElevenLabs_Roger_02.mp3`. Its original export name contains `2026-10-02T00_30_46` and the same Roger voice description. The file is 946607 bytes and 58.122438 seconds long. It is approximately 2.25 seconds longer than recording 01; duration alone cannot explain whether the difference comes from the script, pauses, settings, or another cause.

Both files are mono MP3 audio at 44100 Hz with a nominal audio-stream bitrate of 128 kbps. Both decoded successfully with FFmpeg without reported decode errors. Both satisfy the requested approximate 30-second to two-minute duration. Successful decoding does not establish clear pronunciation, naturalness, factual accuracy, or comfortable loudness.

The packaged MP3 copies retain the exact uploaded bytes; only their filenames are changed to identify synthetic media. The manifest in `results/elevenlabs_artifact_manifest.json` records original filenames, sizes, durations, hashes, codec details, and decode results. The original filenames contain a timestamp-like string; its timezone and the underlying generation time have not been independently verified.

Source text and process record

The existing Task 5 advisory narrative and the earlier Flite source scripts remain available as background. They are not certified transcripts of the new ElevenLabs audio. The exact text entered for each ElevenLabs attempt is needed to check whether the analytical claims and qualifications were preserved and whether the source meets the assignment's roughly 200 to 500 word guidance.

Both export names contain `pre_sp100_s50_sb75_v4`. These strings are recorded as filename evidence, not decoded into verified settings or a model version. The ElevenLabs interface settings, selected model, account tier, free-tier limits, credit use, actual time spent, and any unsuccessful attempts were not supplied with the recordings. No claim of a free or paid generation, particular model version, or safety-filter outcome is made.

A peer can reproduce the broad workflow by opening ElevenLabs, choosing the Roger voice indicated by the exports, entering the confirmed script, recording the actual settings and model, generating two documented iterations, and exporting MP3 files. Exact reproduction requires the missing input text and settings. The accompanying process log separates observed evidence from items still to record.

Comparison across approaches

The project contains two tool families: the earlier local Flite baseline and the student-generated ElevenLabs recordings. Flite approach A used a continuous `kal16` rendering; approach B used `slt`, paragraph segmentation, a tempo factor of 1.12, and inserted pauses. Those baseline commands and outputs are retained for inspection.

The two new Roger exports establish two output files, but their names alone do not establish substantially different prompts, settings, or pipelines. They should not be described as two distinct approaches without a record of what changed. The cross-tool Flite and ElevenLabs comparison supplies a documented difference in production method, while the script and settings gaps prevent a controlled quality comparison. The earlier Flite work was assistant-generated in this session; it is not presented as independent hands-on ElevenLabs work by the student.

Detection result and evidence limits

The attached screenshot is reproduced in the evidence appendix. It displays an AI score of 95 percent, a maximum score of 98 percent, an average score of 95 percent, and 10 analyzed segments. The verdict breakdown assigns all 10 segments to AI Generated, with zero in Likely AI, Likely Human, and Human. The first three fully visible segment scores are 94, 96, and 96 percent.

The screenshot header shows a truncated ElevenLabs filename beginning with the `00_3` portion of the export time, a duration of 0:58, and a size of 924.4 KB. Recording 02 is 58.12 seconds and approximately 924.4 KiB, which strongly suggests that it is the tested file. The screenshot does not expose a full filename or a hash, so this association remains an inference. There is no separate detection result for recording 01.

The detector's provider, model version, threshold, and test URL are not visible in the screenshot. Its explanation refers generally to known AI voice-synthesis patterns; it does not provide enough methodological detail to audit the score. A 95 percent AI score is not a measured 95 percent accuracy rate or a validated probability that the recording is synthetic. Ten segments from one file are not ten independent test recordings. No human control, false-positive test, cross-detector

comparison, or test after re-encoding was supplied.

Disclosure and provenance

All packaged ElevenLabs audio filenames begin `AI_generated`, and the README identifies the files as synthetic speech. The bytes were not edited to insert a spoken disclosure, so the presence and wording of any disclosure already in these recordings must be checked against the audio or an exact transcript before public release. The same check is needed for voice usage rights; a Roger export label is not itself a license or a consent record.

The earlier Flite experiment showed that ordinary MP3 conversion preserved a disclosure comment and an explicitly metadata-stripping conversion removed it. That test was not performed on the ElevenLabs files and does not establish whether an ElevenLabs watermark is present or survives transformations. The metadata inspected in the new MP3s exposes an encoder tag but does not authenticate the creator. No signed C2PA validation or watermark verification has been performed on these exports. Hashes identify preserved bytes, not authorship or consent.

Critical evaluation and remaining reflection

The available technical evidence supports bounded findings: both files are playable, both fit the duration target, and the supplied detector identifies the likely second file as synthetic. It does not establish that either voice would fool a listener or which recording has better cadence, prosody, pronunciation, emotional register, or audibility. No new listener preference or timestamped defects have been reported for this pair.

My earlier comment that the second version was clearer and had distinct, audible words concerned the original Flite comparison. It is retained in `listener_review.md` as historical feedback and is not transferred to the Roger recordings. A final listening review should identify an example from each new file, check names and numerical claims against the source script, and explain whether I would recognize it as synthetic without the label. Lip-sync, blinking, and hand artifacts do not apply to these audio-only deliverables.

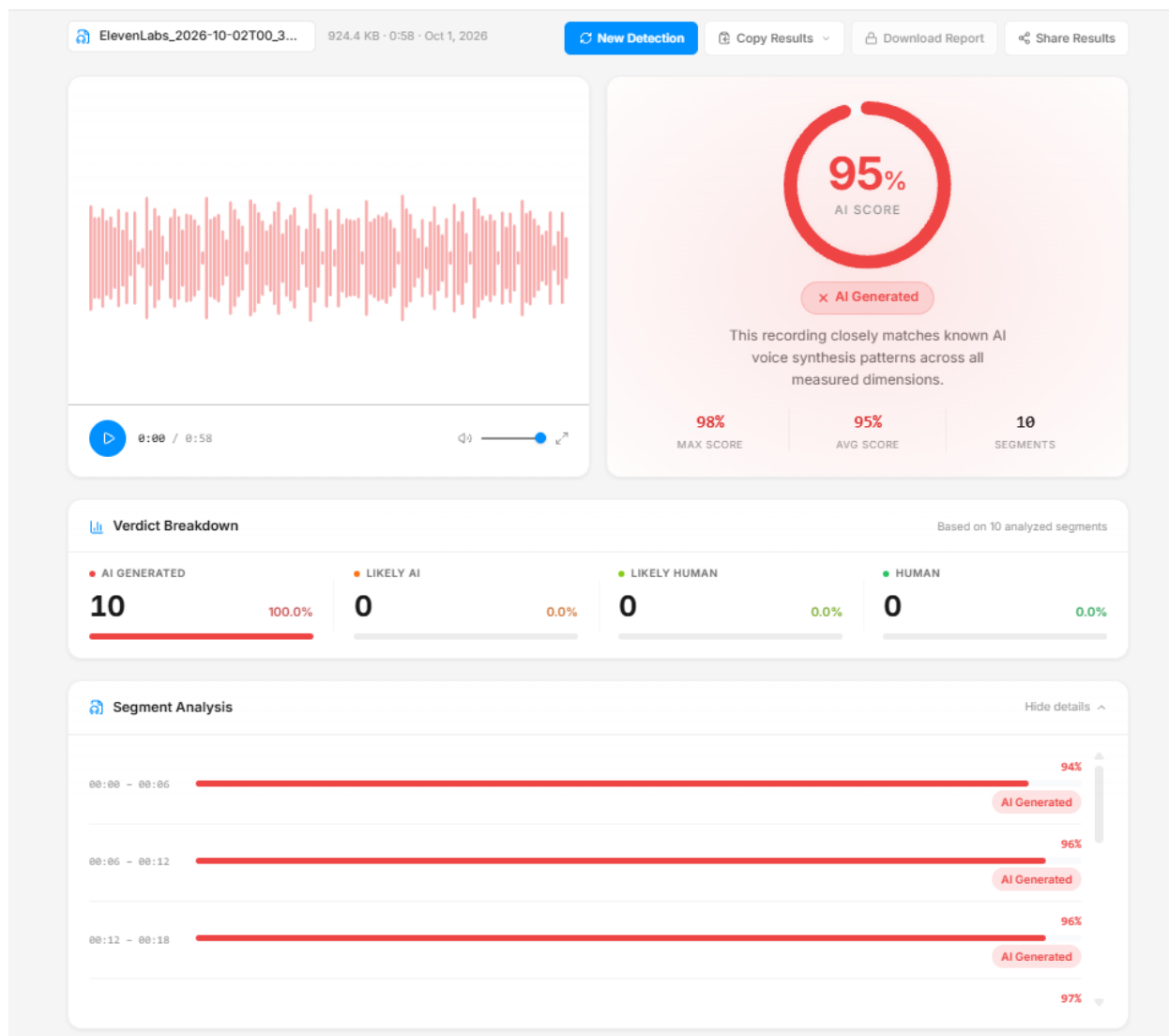
The assignment values honest iteration more than polish. The remaining documentation is the exact source text, the change between the two Roger attempts, actual settings and resource use, and my listening observations. No failed attempts, refusal events, working hours, or emotional reactions have been invented. The earlier chart video remains a Flite-based supplementary artifact; it has not been updated with these recordings and no avatar video has been generated.

Evidence and references

The two student-supplied ElevenLabs exports and the attached detector screenshot are the primary evidence. The original instructor Task 6 brief defines the duration, iteration, disclosure, and documentation requirements. The earlier baseline scripts, command log, and historical report are retained in the full research package. The Task 5 advisory narrative supplies background analysis, but has not been verified as the exact input to the new recordings. The public repository URL was not available when this report was prepared.

Appendix A Detector evidence

Screenshot supplied by the creator. The duration and displayed file size indicate that recording 02 is the likely input; the truncated filename prevents definitive confirmation.



Interpretation: The screenshot reports 95% AI score, 98% maximum score, 95% average score, and 10 of 10 segments classified AI Generated. It supplies no measured error rate, human control, or independent validation. The detector provider is not visible.

Evidence source: user-supplied report Screenshot 2026-10-01 221702.png. The image is reproduced without altering its reported results.